CHE655
LAB 12

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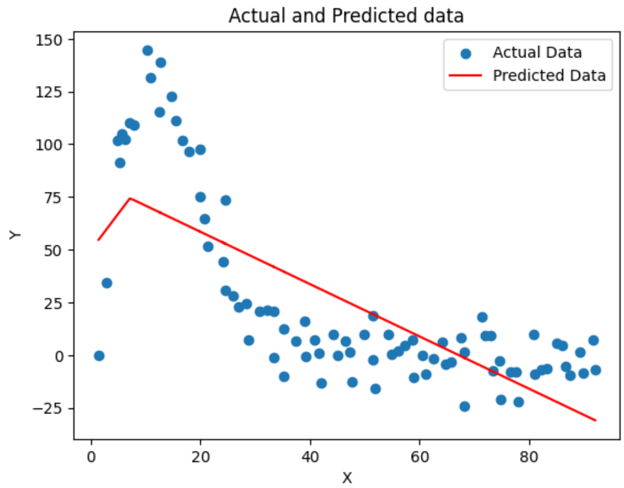
METHODOLOGY

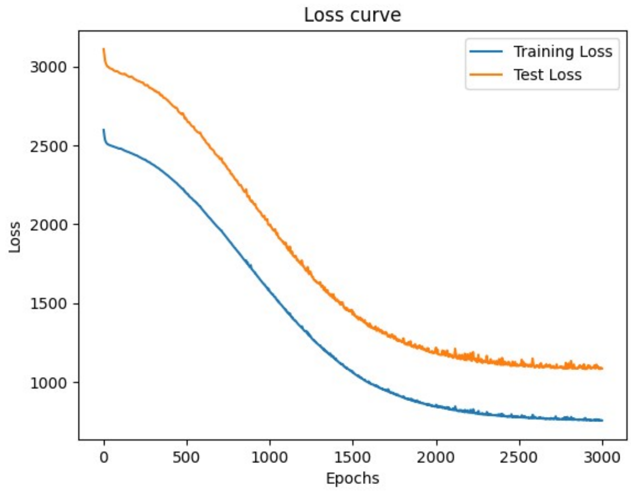
- Data was extracted and trained with 80:20 fit using random_state of 42.
- Then we approximate the neural network using tanh and relu activation functions and different number of layers and different number of neurons per layer.
- The results were then compared.

RESULTS AND OBSERVATIONS

1 hidden layer of 10 neurons , optimisation technique: ADAM and loss function: MSE : -

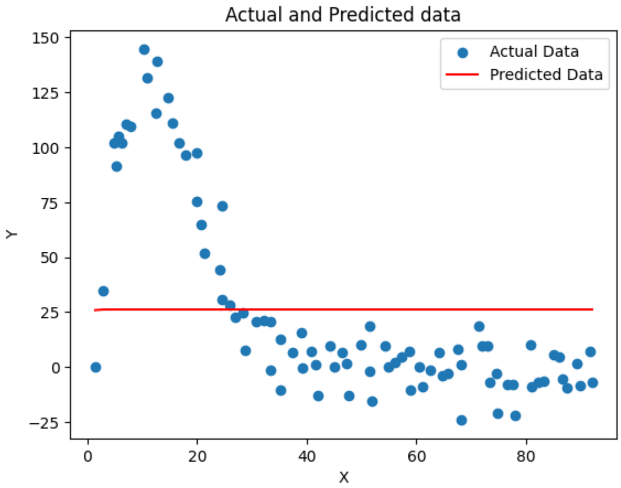
<u>Using ReLU activation function : -</u>

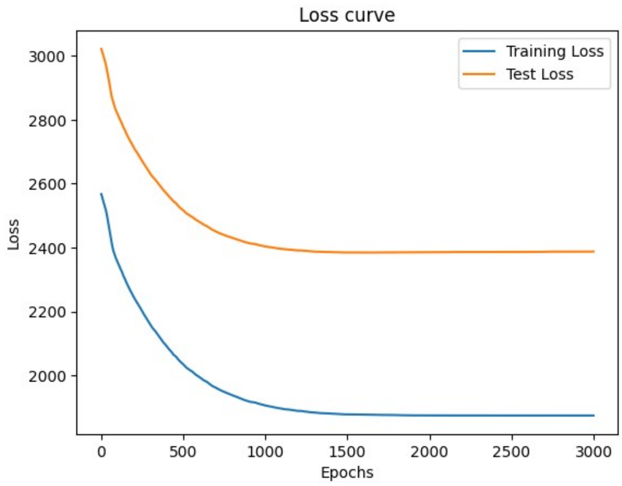




After 3000 epochs, Training data loss settles around 755 while validation data loss settles around 1082

<u>Using tanh activation function : -</u>



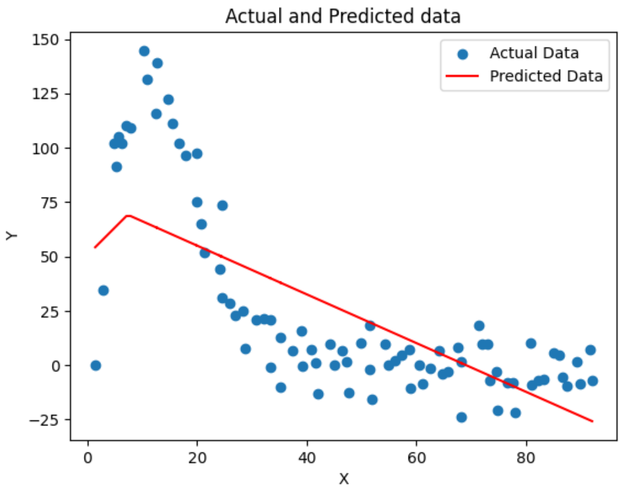


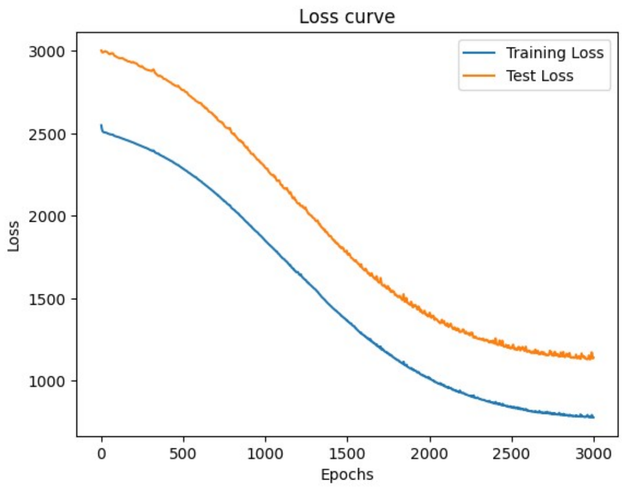
After 3000 epochs, Training data loss settles around 1875 while validation data loss settles around 2387

For 1 hidden layer with 10 neurons, ReLU appears to be the better choice of activation function

1 hidden layer of 5 neurons , optimisation technique: ADAM and loss function: MSE : -

<u>Using ReLU activation function : -</u>

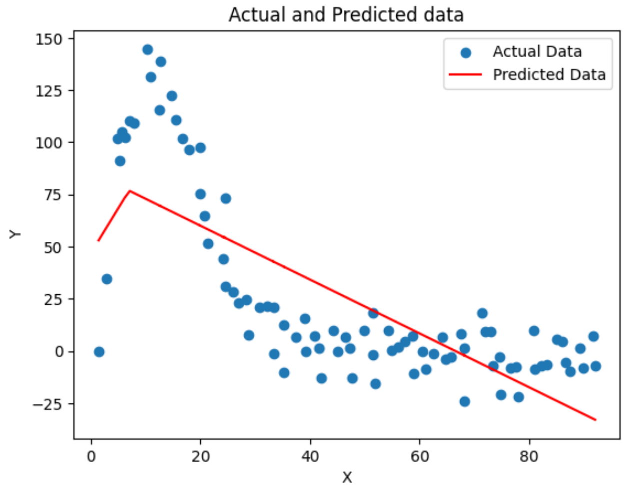


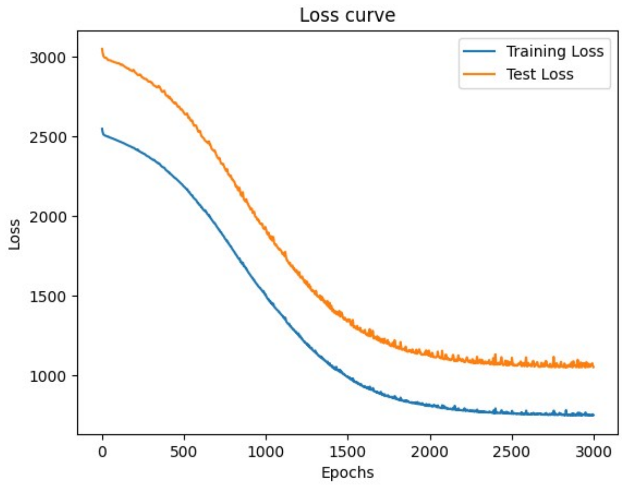


After 3000 epochs, Training data loss settles around 777 while validation data loss settles around 1140

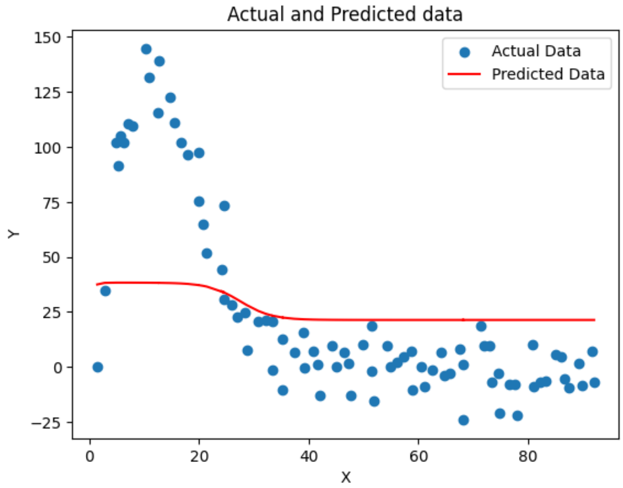
1 hidden layer of 20 neurons , optimisation technique: ADAM and loss function: MSE : -

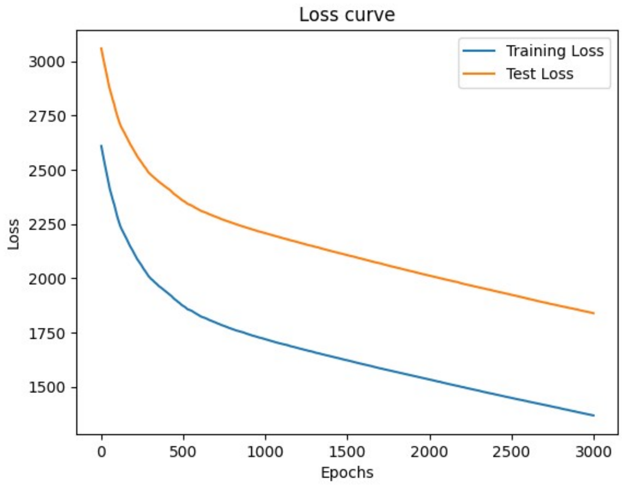
<u>Using ReLU activation function : -</u>





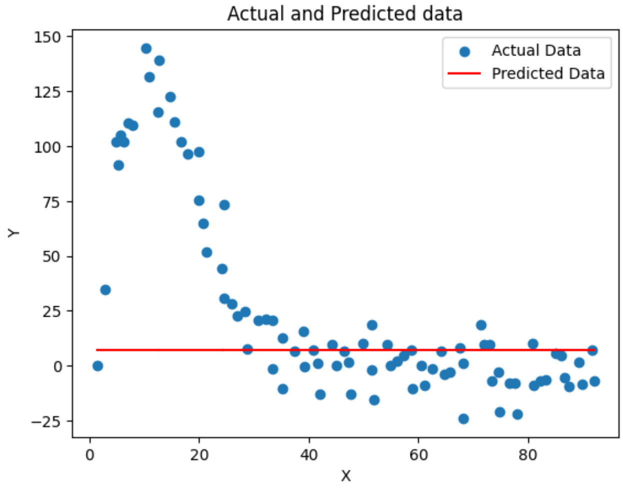
<u>Using tanh activation function : -</u>

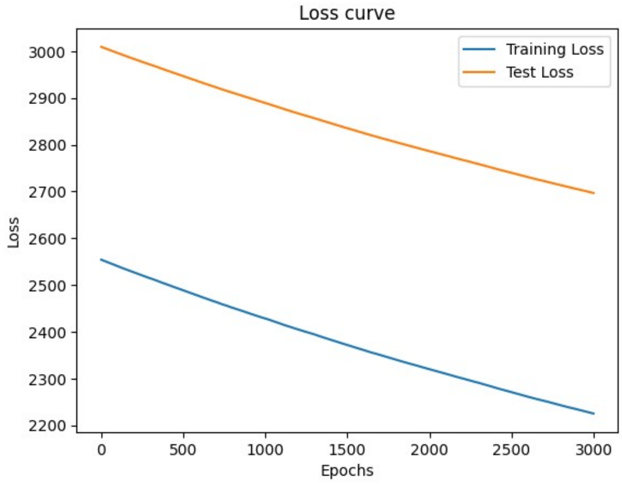




2 hidden layers of 5 neurons followed by 2 neurons , optimisation technique: ADAM and loss function: MSE : -

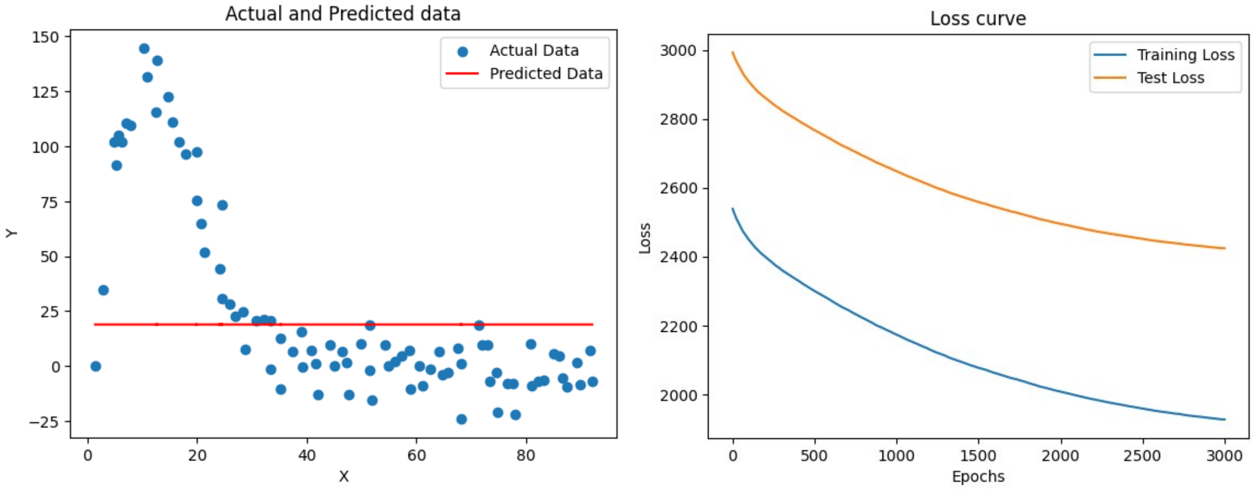
Using ReLU activation function : -





After 3000 epochs, Training data loss settles around 2225 while validation data loss settles around 2836.

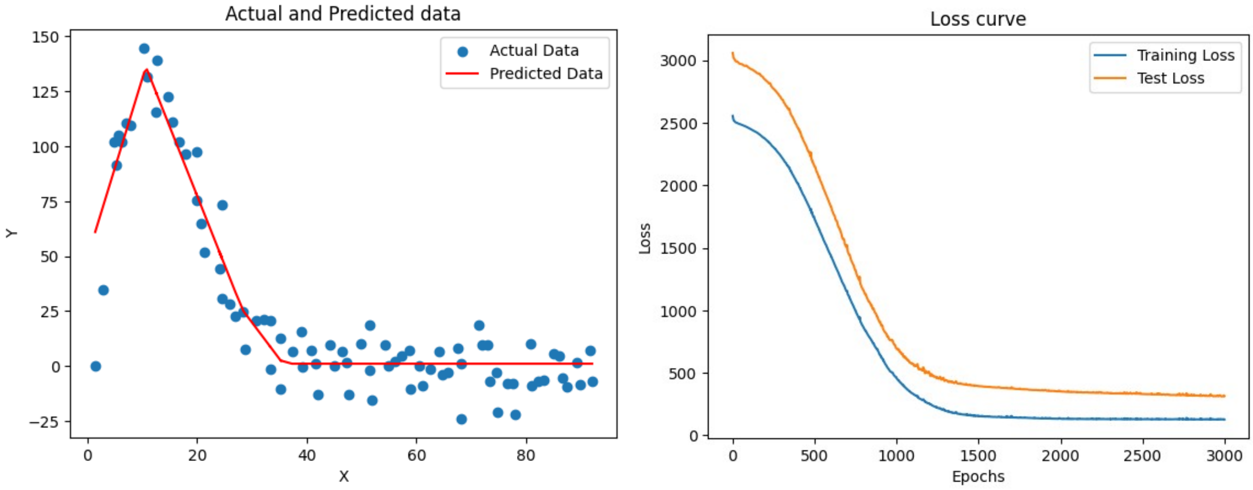
Using tanh activation function : -



After 3000 epochs, Training data loss settles around 1926 while validation data loss settles around 2424.

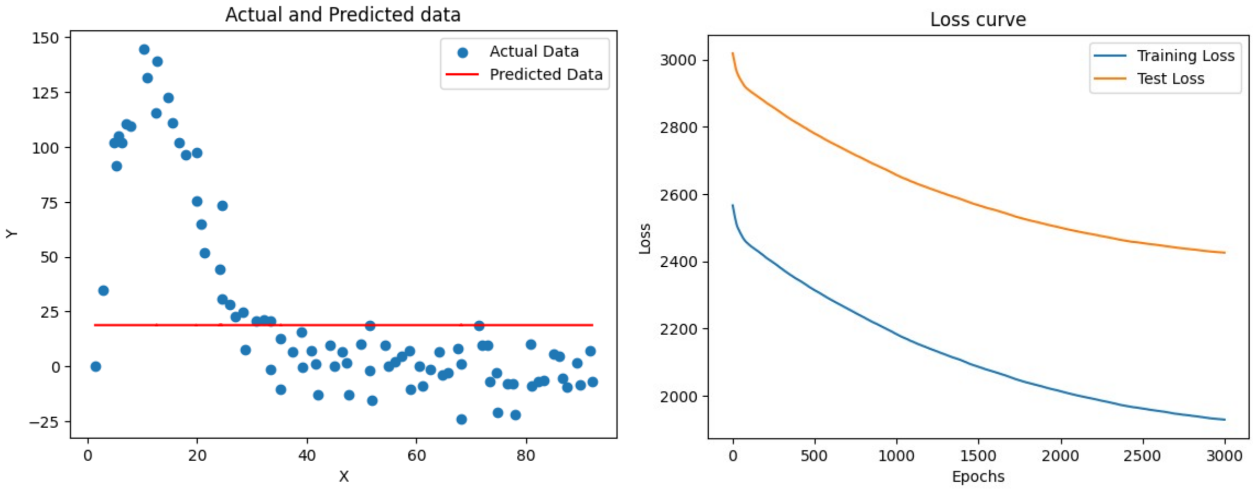
2 hidden layers of 10 neurons followed by 2 neurons , optimisation technique: ADAM and loss function: MSE : -

Using ReLU activation function : -



After 3000 epochs, Training data loss settles around 131 while validation data loss settles around 480.

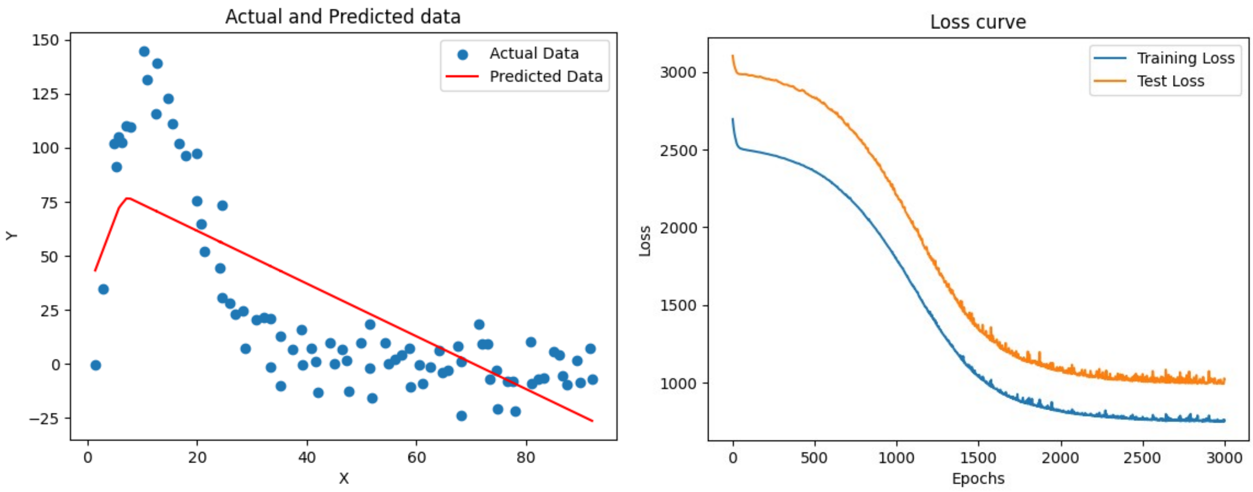
Using tanh activation function : -



After 3000 epochs, Training data loss settles around 1928 while validation data loss settles around 2425.

2 hidden layers of 5 neurons followed by 4 neurons , optimisation technique: ADAM and loss function: MSE : -

<u>Using ReLU activation function : -</u>



After 2500 epochs, Training data loss settles around 760 while validation data loss settles around 1015.

<u>**BEST CHOICE: -**</u>

2 Layers: 10 neuron on layer 1, 2 neurons on layer 2
Activation function: ReLU
optimisation technique: ADAM
loss function: MSE

<u>APPENDIX</u>

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
```

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
import pandas as pd

data = pd.read_csv('/content/Copy of dataset_lab12.csv').values
X = data[:,0]
Y = data[:,1]

X = X.reshape(-1,1)
Y = Y.reshape(-1,1)

X_train,X_test,Y_train,Y_test =
train_test_split(X,Y,test_size=0.2,random_state=42)

model = Sequential([
Dense(5,activation='relu',input_shape=(1,)),
Dense(4,activation='relu'),
Dense(1)
])

model.compile(optimizer='adam',loss='mse')

history =
model.fit(X_train,Y_train,epochs=3000,validation_data=(X_test,Y_test))

Y_pred = model.predict(X)

plt.scatter(X,Y,label = "Actual Data")
plt.plot(X,Y_pred,color='red',label = "Predicted Data")
plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.title("Actual and Predicted data")
plt.show()

plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Test Loss')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.title("Loss curve")
plt.show()
```